

ADSP 31021 Machine Learning Operations

Assignment #3 - AutoML

Due Date - Beginning of Session #6

Assignment Overview

The goal of this assignment is to provide hands-on experience using AutoML within an MLOps workflow. Students will use the same athletes dataset to compare automated model selection, feature importance, model rankings, validation performance, and execution speed across different AutoML approaches.

This assignment emphasizes practical evaluation of AutoML systems, including what AutoML automates, what still requires human judgment, and how AutoML results compare with a previously developed baseline model.

Instructions

- Work individually unless otherwise approved.
- Clearly label all sections and tasks in your notebook, scripts, reports, or screenshots.
- Use the same athletes dataset used in the previous assignments.
- Use your chosen MLOps platform for the first AutoML workflow.
- If your chosen platform is no-code or low-code, include screenshots that clearly show the dataset, AutoML run configuration, leaderboard, model metrics, feature importance, and any reported insights.
- Repeat the required AutoML analysis using H2O AutoML.
- Your submitted workflow and report should be reproducible and easy to follow.
- Total: 100 Points

Reproducibility Requirements

- All code-based submissions must be executable from a clean environment.
- Include dependency management files such as requirements.txt or environment.yml where applicable.
- Provide clear execution instructions for reproducing AutoML runs.
- Use fixed random seed(s) where the platform allows.
- Document train/validation/test split logic or the platform-generated validation strategy.
- Save AutoML outputs, leaderboards, model metrics, feature importance results, and screenshots where applicable.
- Clearly document assumptions, configuration choices, runtime limits, and resource constraints.

Required Tools

- Python 3.x, Jupyter Notebook, or Python scripts for code-based workflows
- Git and GitHub or another submission mechanism approved by the instructor
- Your chosen MLOps platform with AutoML capability
- H2O AutoML: <https://docs.h2o.ai/h2o/latest-stable/h2o-docs/automl.html>
- Libraries required for data loading, preprocessing, model evaluation, visualization, and reporting

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Use the provided dataset: athletes.csv or athletes.csv.zip.

Students should use the same target definition and comparable train/validation/test strategy used in Assignment #1 wherever possible, so AutoML results can be compared fairly against the previously developed model.

Required Workflow Tasks

1. Dataset Loading and Setup (10 Points)

- Load the same athletes dataset used in the previous assignments.
- Document the dataset source, schema, target variable, and any preprocessing applied before AutoML.
- Clearly state whether you are using raw data, cleaned data, or a processed dataset version from Assignment #1.
- Ensure the dataset is ready for AutoML execution.

2. Chosen MLOps Platform AutoML Configuration (10 Points)

- Use your chosen MLOps platform to configure an AutoML run.
- Document the platform name, AutoML settings, target variable, validation strategy, runtime limits, and excluded or included algorithms if applicable.
- If the platform is no-code or low-code, include screenshots of the setup and configuration steps.
- Explain why the selected platform is appropriate for this workflow.

3. AutoML Run Using All Features (10 Points)

- Run AutoML using all appropriate features in the athletes dataset.
- Identify the best model reported by the platform.
- Report the primary validation metric used by the platform.
- Save or capture the AutoML leaderboard and model summary.

4. AutoML Data Insights and Feature Importance (10 Points)

- Report any data insights generated by the AutoML platform.
- Identify and report the top five features from the AutoML run.
- Explain whether the reported insights and feature importance results make sense for the dataset and prediction problem.
- Include screenshots or exported artifacts where applicable.

5. Top Models by Validation Score (10 Points)

- Report the top three models by validation score when using all features.
- Run or configure a second AutoML experiment using only the top features. If you must choose a fixed number of top features, use the top three features.
- Report the top three models by validation score when using only the top features.
- Compare whether feature reduction improved, degraded, or maintained validation performance.

MS Applied Data Science**6. Top Models by Speed (10 Points)**

- Report the top three models by speed when using all features.
- Report the top three models by speed when using only the top features.
- Define how speed was measured, such as training time, prediction time, scoring time, or platform-reported runtime.
- Discuss tradeoffs between validation score and speed.

7. Comparison with Assignment #1 Baseline Model (10 Points)

- Compare the top AutoML models against the model previously developed in Part A of Assignment #1.
- Compare validation score and speed using the most comparable metrics available.
- Discuss whether AutoML improved the model result, reduced development effort, or introduced new complexity.
- Clearly document any limitations in the comparison.

8. Platform AutoML Mode Assessment (10 Points)

- Classify your selected AutoML platform as no-code, low-code, or full-code.
- Explain why you classified it that way.
- Discuss what parts of the workflow were automated and what parts still required manual decisions.
- Comment on operational strengths and risks of using the platform for production-oriented ML workflows.

9. H2O AutoML Repeat (20 Points)

- Repeat the AutoML workflow using H2O AutoML.
- Run H2O AutoML on the athletes dataset and identify the best model.
- Report data insights if available from your H2O workflow.
- Report the top five features from the H2O AutoML workflow.
- Report the top three models by validation score when using all features.
- Compare the H2O AutoML results with your chosen platform AutoML results.

Assignment Requirements

- Use the same athletes dataset as the previous assignments.
- Use your chosen MLOps platform for the primary AutoML workflow.
- Use H2O AutoML for the required repeat workflow.
- Run AutoML using all features and using only the top features.
- Report top models by validation score and by speed.
- Report top five features where available.
- Compare AutoML results to the Assignment #1 baseline model.
- Document screenshots for no-code or low-code workflows.
- Document assumptions, limitations, and configuration choices.

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Deliverables

- GitHub repository, notebook, scripts, or approved Canvas submission containing the working AutoML workflow
- README with setup and execution instructions
- Chosen MLOps platform AutoML configuration and results
- H2O AutoML notebook or script and results
- AutoML leaderboard summaries
- Top five feature importance summary
- Top three models by validation score for all features and top features
- Top three models by speed for all features and top features
- Comparison with Assignment #1 baseline model
- Screenshots for no-code or low-code AutoML platforms
- Short written discussion of insights, tradeoffs, and operational implications